

Arvinth Kumar Kumaravel

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SUMMARY

Experienced Mechanical Design Engineer with 5 years of expertise in CAD, FEA and manufacturing processes. Skilled in 3D modeling, structural analysis, and optimization, with proven expertise in developing innovative designs and streamlining production workflows to improve product performance.

EDUCATION

Purdue University Master of Science in Mechanical Engineering	West Lafayette, USA Aug 2026 - Present
Chalmers University of Technology Master of Science in Product Development, GPA: 3.76/4	Gothenburg, Sweden Jul 2023
Anna University, Coimbatore Institute of Technology Bachelor of Technology in Mechanical Engineering, GPA 3.52/4	TamilNadu, India Apr 2019

TOOLS AND CERTIFICATIONS

Programming	Python, MATLAB, Excel
Design Tools	CATIA, SolidWorks, Siemens NX, Fusion 360, Simufact
Analysis Tools	Ansys, Hypermesh, MSC Adams, Demify
Certifications	AutoCAD, Autodesk Generative Design, SolidWorks

PROFESSIONAL EXPERIENCE

Rototilt Group AB Mechanical Design Engineer	Umeå, Sweden Oct 2023 – Jun 2026
- Designed and developed tilt rotators and excavator implements, improving efficiency of quick couplers by 15% through optimized use of sheet metal, welded constructions, casted components, hydraulics, ensuring CE compliance using Siemens NX. - Conducted Finite Element Analysis (FEM) simulations to ensure structural integrity, alongside prototype development and testing to validate reliability and performance.	
Volvo Cars AB Mechanical Design Engineer	Gothenburg, Sweden Jan 2023 – Jun 2023
- Developed a three-wheeled urban mobility vehicle with a tilting front half to enhance safety and comfort, focusing on innovative design and functionality. - Performed 5+ concept analyses, identified optimal electrification solutions, and developed detailed CATIA models, achieving a 30% increase in structural reliability.	
Hindustan Zinc Limited Mechanical Engineer	Rajasthan, India Jun 2019 – Jun 2021
- Optimized the design of couplings and fasteners for SAG mills, ball mills, and pumps, reducing failure rates by 20% and managing CAD files of spare parts while implementing process optimization techniques. - Led development projects by collaborating with suppliers and internal teams, conducted Failure Mode and Effect Analysis (FMEA) to enhance component quality, and drove Six Sigma methods to foster continuous improvement.	

PROJECTS

- Design of Rescue Boat Towing Arrangement** - Created requirement specifications and conducted in-depth functional and market analyses for a new system. Generated and evaluated ideas using Pugh and Kesselring matrices, then designed and optimized a towing arrangement with CATIA and Ansys, resulting in a full-scale prototype with multiple innovative features.
- Design enhancement of conveyors** - Applied Six Sigma tools, including Cause and Effect Diagrams, Pareto Charts, FMEA, Stratification, and Affinity Diagrams, to identify and analyze root causes of spillage. Designed, developed, and implemented technical solutions such as PU scrapers, PU skirts, self-aligning frames, and wash boxes, resulting in a 30% reduction in spillage.

LEADERSHIP EXPERIENCE

Communications Leader, Chalmers International Receptions Committee	Nov 2021 – Dec 2022
Event Manager, CHARM Event, Chalmers University of Technology	June 2022 – Sep 2022

SCHOLARSHIPS

Rotary International Student House Scholarship (1000 USD)	Nov 2022
International Programme Office for Education and Training (IPOET) Scholarship (25000 USD)	Apr 2021